$$\frac{m}{m}\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

know $e^{rt} = x$ r=constant

$$\frac{dx}{dt} = re^{rt}, \quad \frac{d^2x}{dt^2} = r^2 e^{rt}$$

$$r^2 e^{rt} + \frac{k}{m} e^{rt} = 0$$

$$\Rightarrow e^{rt}\left(r^2 + \frac{k}{m}\right) = 0 \iff r^2 = -\frac{k}{m}$$

since $\frac{k}{m} > 0$ let $\frac{k}{m} = \omega^2$

$$\Rightarrow r^2 = -\omega^2$$

Take square root: $r = \pm\sqrt{-\omega^2} = \pm j\omega$ where $\omega = \sqrt{\frac{k}{m}}$

Solution: $x(t) = ae^{+j\omega t} + be^{-j\omega t}$

Note: $e^{\pm j\omega t} = \cos(\omega t) \pm j\sin(\omega t)$

Rewrite solution as: $x(t) = A\cos(\omega t) + B\sin(\omega t)$

Check: if $x = A\cos(\omega t) \Rightarrow \frac{d^2x}{dt^2} = -\omega^2 \underbrace{A\cos(\omega t)}_{x} \Rightarrow -\omega^2 x + \omega^2 x = 0$ ✓

Start with $x(t) = \left[a\cos(\omega t) + b\sin(\omega t)\right]\frac{\sqrt{a^2+b^2}}{\sqrt{a^2+b^2}}$

notice that $\left[\frac{a}{\sqrt{a^2+b^2}}\right] < 1$ and $\left[\frac{b}{\sqrt{a^2+b^2}}\right] < 1$

let $x(t) = \sqrt{a^2+b^2}\left[\frac{a}{\sqrt{a^2+b^2}}\cos(\omega t) + \frac{b}{\sqrt{a^2+b^2}}\sin(\omega t)\right]$

Now note: $\left(\frac{a}{\sqrt{a^2+b^2}}\right)^2 + \left(\frac{b}{\sqrt{a^2+b^2}}\right)^2 = 1 \Rightarrow$ let $\frac{a}{\sqrt{a^2+b^2}} = \cos\phi_o$

$\frac{b}{\sqrt{a^2+b^2}} = \sin\phi_o$

Now $x(t) = A\left[\cos(\omega t)\cos(\phi_o) + \sin(\omega t)\sin(\phi_o)\right]$

$\cos A\cos B + \sin A\sin B = \cos(A-B)$

$\Rightarrow \boxed{x(t) = A\cos(\omega t + \phi_o)}$ equivalent way to express the solution.